

Unit 10: Product Rule & Quotient Rule — Full Solutions

Warm-up

1. $y = (x^2)(x+3)$

Let $f = x^2$ (so $f' = 2x$) and $g = x+3$ (so $g' = 1$).

$$y' = f'g + fg' = 2x(x+3) + x^2(1) = 2x^2 + 6x + x^2 = 3x^2 + 6x$$

$y' = 3x^2 + 6x$. (Double-check by expanding first: $x^2(x+3) = x^3 + 3x^2$, and its derivative is $3x^2 + 6x$ — matches!)

2. $y = (2x+1)(x-4)$

$f = 2x+1$ ($f' = 2$), $g = x-4$ ($g' = 1$).

$$y' = 2(x-4) + (2x+1)(1) = 2x - 8 + 2x + 1 = 4x - 7$$

$$y' = 4x - 7.$$

3. $y = (x^2+1)(x-2)$

$f = x^2+1$ ($f' = 2x$), $g = x-2$ ($g' = 1$).

$$y' = 2x(x-2) + (x^2+1)(1) = 2x^2 - 4x + x^2 + 1 = 3x^2 - 4x + 1$$

$$y' = 3x^2 - 4x + 1.$$

4. $y = \frac{x+1}{x-1}$

$f = x+1$ ($f' = 1$), $g = x-1$ ($g' = 1$).

$$y' = \frac{1 \cdot (x-1) - (x+1) \cdot 1}{(x-1)^2} = \frac{(x-1) - (x+1)}{(x-1)^2} = \frac{-2}{(x-1)^2}$$

$$y' = -\frac{2}{(x-1)^2}.$$

5. $y = \frac{3x-2}{x+5}$

$f = 3x-2$ ($f' = 3$), $g = x+5$ ($g' = 1$).

$$y' = \frac{3(x+5) - (3x-2)(1)}{(x+5)^2} = \frac{3x+15-3x+2}{(x+5)^2} = \frac{17}{(x+5)^2}$$

$$y' = \frac{17}{(x+5)^2}.$$

6. $y = \frac{x^2}{x+1}$

$$f = x^2 \ (f' = 2x), \ g = x + 1 \ (g' = 1).$$

$$y' = \frac{2x(x+1) - x^2(1)}{(x+1)^2} = \frac{2x^2 + 2x - x^2}{(x+1)^2} = \frac{x^2 + 2x}{(x+1)^2}$$

$$y' = \frac{x^2 + 2x}{(x+1)^2}.$$

$$7. \ y = (x-3)(x^2+2)$$

$$f = x-3 \ (f' = 1), \ g = x^2+2 \ (g' = 2x).$$

$$y' = 1 \cdot (x^2+2) + (x-3)(2x) = x^2+2+2x^2-6x = 3x^2-6x+2$$

$$y' = 3x^2 - 6x + 2.$$

Standard

$$8. \ y = (x^2+1) \left(x+1+\frac{1}{x} \right)$$

Rewrite the second factor using powers: $g = x+1+x^{-1}$, so $g' = 1-x^{-2}$.

Let $f = x^2+1 \ (f' = 2x)$.

$$y' = 2x \left(x+1+\frac{1}{x} \right) + (x^2+1) \left(1-\frac{1}{x^2} \right)$$

Expand the first piece: $2x \cdot x + 2x \cdot 1 + 2x \cdot \frac{1}{x} = 2x^2 + 2x + 2$.

Expand the second piece: $(x^2+1)(1) - (x^2+1) \cdot \frac{1}{x^2} = x^2+1 - (1+\frac{1}{x^2}) = x^2+1-1-\frac{1}{x^2} = x^2 - \frac{1}{x^2}$.

Add them together:

$$y' = (2x^2 + 2x + 2) + \left(x^2 - \frac{1}{x^2} \right) = 3x^2 + 2x + 2 - \frac{1}{x^2}$$

$$y' = 3x^2 + 2x + 2 - \frac{1}{x^2}.$$

$$9. \ y = \frac{x+5}{3x+2}$$

$$f = x+5 \ (f' = 1), \ g = 3x+2 \ (g' = 3).$$

$$y' = \frac{1 \cdot (3x+2) - (x+5)(3)}{(3x+2)^2} = \frac{3x+2-3x-15}{(3x+2)^2} = \frac{-13}{(3x+2)^2}$$

$$y' = -\frac{13}{(3x+2)^2}.$$

10. $y = \frac{x^2 - 3}{2x + 1}$

$f = x^2 - 3$ ($f' = 2x$), $g = 2x + 1$ ($g' = 2$).

$$y' = \frac{2x(2x + 1) - (x^2 - 3)(2)}{(2x + 1)^2} = \frac{4x^2 + 2x - 2x^2 + 6}{(2x + 1)^2} = \frac{2x^2 + 2x + 6}{(2x + 1)^2}$$

$$y' = \frac{2x^2 + 2x + 6}{(2x + 1)^2}.$$

11. $y = (x + 2)(x^2 - 3x + 1)$

$f = x + 2$ ($f' = 1$), $g = x^2 - 3x + 1$ ($g' = 2x - 3$).

$$y' = 1 \cdot (x^2 - 3x + 1) + (x + 2)(2x - 3)$$

Expand $(x + 2)(2x - 3) = 2x^2 - 3x + 4x - 6 = 2x^2 + x - 6$.

$$y' = (x^2 - 3x + 1) + (2x^2 + x - 6) = 3x^2 - 2x - 5$$

$$y' = 3x^2 - 2x - 5.$$

12. $y = \left(\frac{1}{x}\right)(x^2 + 4)$

Rewrite: $f = x^{-1}$ ($f' = -x^{-2}$), $g = x^2 + 4$ ($g' = 2x$).

$$y' = -x^{-2}(x^2 + 4) + x^{-1}(2x)$$

Simplify each piece: $-x^{-2}(x^2 + 4) = -1 - 4x^{-2}$. And $x^{-1}(2x) = 2$.

$$y' = (-1 - 4x^{-2}) + 2 = 1 - \frac{4}{x^2}$$

$y' = 1 - \frac{4}{x^2}$. (Check: simplifying first, $y = \frac{x^2 + 4}{x} = x + 4x^{-1}$, and $y' = 1 - 4x^{-2}$ — matches!)

13. $y = \frac{2x - 1}{x^2 + 3}$

$f = 2x - 1$ ($f' = 2$), $g = x^2 + 3$ ($g' = 2x$).

$$y' = \frac{2(x^2 + 3) - (2x - 1)(2x)}{(x^2 + 3)^2} = \frac{2x^2 + 6 - (4x^2 - 2x)}{(x^2 + 3)^2} = \frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}$$

$$y' = \frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}.$$

14. $y = (x^2 + x)(3 - x)$

$$f = x^2 + x \ (f' = 2x + 1), \ g = 3 - x \ (g' = -1).$$

$$y' = (2x + 1)(3 - x) + (x^2 + x)(-1)$$

$$\text{Expand } (2x + 1)(3 - x) = 6x - 2x^2 + 3 - x = -2x^2 + 5x + 3.$$

$$y' = (-2x^2 + 5x + 3) - (x^2 + x) = -3x^2 + 4x + 3$$

$$y' = -3x^2 + 4x + 3.$$

Challenge

$$15. \ y = \frac{(x^2 + x)(x^2 - x + 1)}{x^4}$$

This is much easier if we simplify first. Expand the numerator:

$$(x^2 + x)(x^2 - x + 1) = x^2(x^2 - x + 1) + x(x^2 - x + 1)$$

$$= (x^4 - x^3 + x^2) + (x^3 - x^2 + x) = x^4 - x^3 + x^2 + x^3 - x^2 + x$$

Notice the $-x^3$ and $+x^3$ cancel, and the x^2 and $-x^2$ cancel:

$$= x^4 + x$$

So the whole function simplifies dramatically:

$$y = \frac{x^4 + x}{x^4} = 1 + \frac{x}{x^4} = 1 + x^{-3}$$

Now just differentiate:

$$y' = -3x^{-4} = -\frac{3}{x^4}$$

$y' = -\frac{3}{x^4}$. (This is a great example of why checking for simplification first can save you from a much messier quotient-rule computation!)

$$16. \ y = \sqrt{x}(x^2 - 1)$$

Rewrite: $f = x^{1/2}$ ($f' = \frac{1}{2}x^{-1/2}$), $g = x^2 - 1$ ($g' = 2x$).

$$y' = \frac{1}{2}x^{-1/2}(x^2 - 1) + x^{1/2}(2x)$$

Write both pieces over a common denominator of $2\sqrt{x}$: the first piece is already $\frac{x^2-1}{2\sqrt{x}}$. The second piece, $2x\sqrt{x} = 2x^{3/2}$, can be rewritten as $\frac{4x^2}{2\sqrt{x}}$ (since $\frac{4x^2}{2\sqrt{x}} = 2x^2x^{-1/2} = 2x^{3/2}$).

$$y' = \frac{(x^2-1)+4x^2}{2\sqrt{x}} = \frac{5x^2-1}{2\sqrt{x}}$$

$$y' = \frac{5x^2-1}{2\sqrt{x}}.$$

$$17. y = \frac{x^2-4}{\sqrt{x}}$$

This is much easier if we simplify first by dividing each term by $\sqrt{x} = x^{1/2}$:

$$y = \frac{x^2}{x^{1/2}} - \frac{4}{x^{1/2}} = x^{3/2} - 4x^{-1/2}$$

Now differentiate:

$$y' = \frac{3}{2}x^{1/2} - 4\left(-\frac{1}{2}\right)x^{-3/2} = \frac{3}{2}x^{1/2} + 2x^{-3/2}$$

$y' = \frac{3}{2}\sqrt{x} + \frac{2}{x^{3/2}}$. (Again, simplifying before differentiating was far cleaner than forcing the quotient rule here.)

$$18. y = \left(x + \frac{1}{x}\right)(x^2 - 2)$$

Rewrite: $f = x + x^{-1}$ ($f' = 1 - x^{-2}$), $g = x^2 - 2$ ($g' = 2x$).

$$y' = (1 - x^{-2})(x^2 - 2) + (x + x^{-1})(2x)$$

Expand the first piece: $(1)(x^2 - 2) - x^{-2}(x^2 - 2) = (x^2 - 2) - (1 - 2x^{-2}) = x^2 - 2 - 1 + 2x^{-2} = x^2 - 3 + 2x^{-2}$.

Expand the second piece: $2x \cdot x + 2x \cdot x^{-1} = 2x^2 + 2$.

Add them together:

$$y' = (x^2 - 3 + 2x^{-2}) + (2x^2 + 2) = 3x^2 - 1 + \frac{2}{x^2}$$

$$y' = 3x^2 - 1 + \frac{2}{x^2}.$$

$$19. y = \frac{x^3-1}{x^2+1}$$

$f = x^3 - 1$ ($f' = 3x^2$), $g = x^2 + 1$ ($g' = 2x$).

$$y' = \frac{3x^2(x^2+1) - (x^3-1)(2x)}{(x^2+1)^2} = \frac{3x^4+3x^2-(2x^4-2x)}{(x^2+1)^2} = \frac{x^4+3x^2+2x}{(x^2+1)^2}$$

$$y' = \frac{x^4+3x^2+2x}{(x^2+1)^2}.$$

20. Find the tangent line to $y = (2x-1)(x+3)$ at $x = 1$.

First, find the point: $y(1) = (2(1)-1)(1+3) = (1)(4) = 4$. So the point is $(1,4)$.

Next, find the derivative. Let $f = 2x-1$ ($f' = 2$), $g = x+3$ ($g' = 1$).

$$y' = 2(x+3) + (2x-1)(1) = 2x+6+2x-1 = 4x+5$$

Evaluate at $x = 1$: $y'(1) = 4(1) + 5 = 9$.

Using point-slope form with point $(1,4)$ and slope 9:

$$y-4 = 9(x-1)$$

$$y = 9x - 9 + 4 = 9x - 5$$

Tangent line: $y = 9x - 5$.

Applied

21. $R(x) = x \cdot (50 - 0.5x)$

Let $f = x$ ($f' = 1$), $g = 50 - 0.5x$ ($g' = -0.5$).

$$R'(x) = 1 \cdot (50 - 0.5x) + x \cdot (-0.5) = 50 - 0.5x - 0.5x = 50 - x$$

Evaluate at $x = 10$:

$$R'(10) = 50 - 10 = 40$$

Answer: $R'(x) = 50 - x$, and $R'(10) = 40$. At a sales level of 10 units, selling one more unit adds about \$40 to total revenue.

22. $\bar{A}(x) = \frac{C(x)}{x} = \frac{x^2 + 20x + 100}{x}$

This is easier to simplify first by dividing each term by x :

$$\bar{A}(x) = x + 20 + \frac{100}{x} = x + 20 + 100x^{-1}$$

Differentiate:

$$\bar{A}'(x) = 1 - 100x^{-2} = 1 - \frac{100}{x^2}$$

Evaluate at $x = 5$:

$$\bar{A}'(5) = 1 - \frac{100}{25} = 1 - 4 = -3$$

Answer: $\bar{A}'(x) = 1 - \frac{100}{x^2}$, and $\bar{A}'(5) = -3$. At 5 units of production, the average cost per unit is currently decreasing at a rate of \$3 per unit as production increases.

$$23. D(t) = \frac{P(t)}{A(t)} = \frac{t^2 + 5t}{t + 2}$$

$$f = t^2 + 5t \ (f' = 2t + 5), \ g = t + 2 \ (g' = 1).$$

$$D'(t) = \frac{(2t + 5)(t + 2) - (t^2 + 5t)(1)}{(t + 2)^2}$$

$$\text{Expand } (2t + 5)(t + 2) = 2t^2 + 4t + 5t + 10 = 2t^2 + 9t + 10.$$

$$D'(t) = \frac{(2t^2 + 9t + 10) - (t^2 + 5t)}{(t + 2)^2} = \frac{t^2 + 4t + 10}{(t + 2)^2}$$

Evaluate at $t = 3$:

$$D'(3) = \frac{9 + 12 + 10}{(5)^2} = \frac{31}{25}$$

Answer: $D'(t) = \frac{t^2 + 4t + 10}{(t + 2)^2}$, and $D'(3) = \frac{31}{25}$. At year 3, the population density is increasing at a rate of $\frac{31}{25}$ (thousand people per square mile, per year).

$$24. N(x) = (x + 10)(50 - 2x)$$

$$f = x + 10 \ (f' = 1), \ g = 50 - 2x \ (g' = -2).$$

$$N'(x) = 1 \cdot (50 - 2x) + (x + 10)(-2) = 50 - 2x - 2x - 20 = 30 - 4x$$

Evaluate at $x = 5$:

$$N'(5) = 30 - 20 = 10$$

Answer: $N'(x) = 30 - 4x$, and $N'(5) = 10$. At $x = 5$, the store's revenue is increasing at a rate of \$10 for each small increase in x .